WHL3.

Robotic Exoskeleton for Load Transportation

By: Yim Fu Chong (1155176974)

Supervisor: Professor Wei-Hsin Liao

Final Year Project Final Report

Submitted in Partial Fulfillment of the
Requirements for

The Degree of Bachelor of Engineering

Department of Mechanical and Automation Engineering
The Chinese University of Hong Kong

April 2, 2026

Acknowledgments

I would like to express my sincere gratitude to my supervisor Professor Wei-Hsin Liao. His insights into robotic exoskeletons and actuation systems have been instrumental in shaping the direction and progress of my work.

I am also deeply thankful to the team at ExoTechHK Limited for their collaboration and providing access to their lower-limb exoskeleton prototype. This partnership, facilitated under the Hong Kong Science and Technology Parks (HKSTP) incubation programme, has offered a practical pathway for technology transfer and real-world application.

Special thanks go to the Department of Mechanical and Automation Engineering at The Chinese University of Hong Kong for the resources and facilities that enabled this research. Additionally, I appreciate the contributions from fellow researchers and peers who provided feedback during the development phase.

Finally, I extend my appreciation to my family and friends for their encouragement and understanding during this challenging yet rewarding endeavor.

Yim Fu Chong
April 2, 2026

Abstract

Main objectives

This project aims to develop a software-centric enhancement module for a lower-limb robotic exoskeleton designed to assist workers in heavy load transportation tasks (> 20 kg) in industrial, logistics, and construction environments. The primary objectives are to enable real-time detection of user locomotion intent (including walking, stair ascent/descent, sit-to-stand transitions, and speed variations), accurately estimate joint kinematics through multi-sensor fusion, and deliver adaptive personalized torque assistance to minimize physical effort and injury risk. To achieve this, the module will incorporate computational techniques to handle variability in user biomechanics and environmental conditions, ensuring robust performance across diverse scenarios.

Planned methods (interim report, Jan 2026 PDF)

The proposed system integrates inertial measurement units (IMUs) placed on the lower limbs and back, with data processed through a Kalman filter-based sensor fusion pipeline implemented in MATLAB to achieve robust, low-latency joint angle and velocity estimation. Locomotion mode and transition prediction were planned using machine learning models (Support Vector Machines and Long Short-Term Memory networks) trained on labeled gait datasets, targeting $> 90\%$ classification accuracy and a 100–200 ms predictive horizon, with performance evaluated via confusion matrices, precision, and recall. Finite state machines were incorporated for locomotion mode transitions. The data collection protocol involves attaching three mobile phones as IMUs to the hips and lower back; performing standardized tasks such as walking 10 m on flat ground, climbing 10 steps, or navigating uneven surfaces at self-selected speeds; and exporting accelerometer, gyroscope, and magnetometer data as CSV files for processing. Joint kinematics information is intended for precise control.

Implemented methods (repository baseline, updated)

The validated MATLAB baseline uses Kalman fusion (`imufilter`) as above. Classification uses overlapping windows (100 samples at 100 Hz, step 50).

The **primary deployment classifier** is a **binary** RBF SVM (`fitcsvm`, standard-

ized features) with a **30-dimensional** training vector: five statistics per IMU slot × six slots (`Features.m`); HuGaDB supplies six on-body IMUs per window; USC-HAD and on-line simulation use one IMU stream with remaining slots zero-padded (`LocomotionFeatureVector`). The positive class groups **locomotion-like** activities (level walk, stairs, run per dataset label maps).

A **multiclass** ECOC ensemble (`fitcecoc`) provides twelve unified activity labels for analysis; exoskeleton gating uses the binary model and an asymmetric FSM (3 consecutive walk-like predictions to assist, 5 to release).

An **optional binary LSTM** (Deep Learning Toolbox) classifies the same locomotion-like vs. non-locomotion task from **sequence** inputs (raw IMU windows mapped to feature×time matrices) via `TrainLstmBinary` and `RunExoskeletonPipelineLstm`; it is an alternative to the RBF SVM, not a hybrid SVM+LSTM fusion stage.

The 100–200 ms horizon remains a *design target*; literature reports comparable latencies for kinematic lift-onset cues [1], but the submitted baselines were not benchmarked under that specific end-to-end metric.

Interim results statement (Dec 2025 / Jan 2026 PDF)

The interim submission reported: a functional three-phone acquisition prototype; preliminary Kalman filtering and intention-detection algorithms validated on public datasets including HuGaDB and the USC-SIPI gait database; repository setup; real-time visualization; and initial locomotion classification with **98.88%** accuracy and a **150 ms** predictive horizon. *Corrections:* the smartphone gait database used in this project is **USC-HAD** [2], not USC-SIPI. The 150 ms figure overlapped claims from the literature on hip-based lift detection [1] and was not rigorously established as an end-to-end latency of the final classifier at submission time.

Updated empirical results (merged public data, repository metrics)

Kalman filtering and SVM-based intention detection are implemented in MATLAB. Training uses **USC-HAD** [2] and **HuGaDB** [3] with merged windows and the 30-D feature layout. **Binary** five-fold out-of-fold accuracy from the committed evaluation files is **98.91%** (USC-HAD only), **88.60%** (HuGaDB only), and **94.50%** (merged); the default saved binary model is merged. The twelve-class ECOC model achieves **56.76%** OOF accuracy on a stratified subsample used for tractable cross-validation (see `results/multiclass_evaluation_metrics.mat`). Real-time pipeline simulation on CSV data shows fused hip flexion angle and FSM-latched assist commands. Scalar accuracies in this report are regenerated into `docs/latex/generated_metrics.tex` by `scripts/ExportMetricsForReport.m`. Optional LSTM training yields a separate 20%

holdout metric (see Results; not comparable to five-fold SVM OOF) and may omit the confusion figure if training is interrupted before model export.

Main conclusions

This project demonstrates that automation technology, particularly machine-learning-based intention recognition combined with sensor fusion and finite state machines, is applicable to next-generation powered exoskeleton control supervisors. When fully realized, the proposed module has strong potential for integration into industrial platforms (e.g., ExoTechHK Limited), offering high accuracy and balanced metrics on public IMU data while enhancing safety through hysteresis-based gating. This work contributes to workplace ergonomics and provides a reproducible foundation for hardware-in-the-loop studies. The repository now documents both the **original project plan** (hybrid SVM+LSTM fusion, extended fusion) and the **delivered MATLAB baseline** (binary + optional multiclass SVM, dual-dataset training, optional binary LSTM sequence classifier).

Contents

Acknowledgments **i**

Abstract **ii**

1 Problem Description **1**
1.1 Problem Statement 1
1.2 Project Goal 1
1.3 Significance of the Project 2

2 Background Knowledge and Literature Review **3**
2.1 Introduction 3
2.2 Overview of back-support exoskeletons for MMH 3
2.3 Actuation Systems (Cable-Driven and SEAs) 5
2.4 Intention Detection, Sensor Fusion, and Machine Learning 5
2.5 Evaluation Studies and User Acceptance 6
2.6 Gaps and Implications for the Project 6
2.7 Representative Exoskeletons (Summary Table) 7

3 Proposed Solution Methods **8**
3.1 Proposed Methodologies 8
3.2 Intention Detection Algorithm (planned and implemented) 8
3.3 Multi-Sensor Data Processing Pipeline (planned and implemented) 9
3.4 Machine Learning Model Training and Deployment (planned and implemented) 10
3.5 System Integration and Control (planned and implemented) 11
3.6 List of MATLAB Libraries and Toolboxes 11

4 Results and Discussions **12**
4.1 Experimental Outputs Used in This Chapter 12
4.2 Binary Locomotion Recognition Results 12
4.3 Discussion of Dataset Effects 13
4.4 Multiclass Activity Recognition 14
4.5 Real-Time Pipeline Behavior 15

4.6 Sequence-Model Status 15
4.7 Overall Interpretation 17
4.8 Visualization Tools and Software Architecture 17

5 Conclusions and Future Work **18**
5.1 Summary of what was delivered 18
5.2 Future work beyond the FYP (interim PDF) 18
5.3 Project timeline (interim PDF) 19
5.4 Risk assessment (interim PDF) 19
5.5 Repository status (consolidated, 2026) 19

References **21**

Chapter 1

Problem Description

1.1 Problem Statement

Manual load transportation in industrial, construction, and logistics settings frequently involves carrying loads exceeding 20 kg, surpassing human physiological limits and leading to high rates of lower-back pain, musculoskeletal disorders (MSDs), reduced productivity, and billions in annual economic costs worldwide. Existing assistive exoskeletons often fail to provide proactive, real-time adaptation to the user's locomotion intent or environmental changes, resulting in delayed or mismatched assistance that compromises safety and comfort. Moreover, generic assistance profiles do not account for inter-individual differences in gait asymmetry, fatigue thresholds, anthropometrics, or preferred movement patterns. These limitations are especially critical in dynamic workplaces where load magnitude, terrain, and tasks constantly vary. There remains a clear need for a lightweight, cost-effective lower-limb exoskeleton that delivers accurate intent detection, robust sensor fusion, and truly personalized adaptive control—without relying on invasive interfaces such as electromyography (EMG) or requiring extensive user training.

1.2 Project Goal

The primary goal is to develop a software-centric enhancement module for ExoTechHK Limited's existing lower-limb exoskeleton prototype—a local startup incubated at HKSTP and founded by CUHK researchers—to augment human strength and endurance during heavy load-carrying tasks (> 20 kg) without invasive sensors or lengthy calibration.

Original project goal statement (interim PDF, Jan 2026)

The module was specified to integrate: (i) a **data acquisition and processing pipeline** using three mobile phones as IMUs, Kalman-based fusion for joint kinematics, and *initial* machine learning (e.g., hybrid SVM+LSTM) for locomotion classification with validation

on public datasets; and (ii) **proactive real-time locomotion intent recognition** using hip and lower-back IMU data via a hybrid SVM+LSTM model with multi-sensor fusion, targeting $> 90\%$ intent-detection accuracy and a 100–200 ms predictive horizon, with evaluation through confusion matrices.

Current implementation emphasis (repository)

The module integrates two core components as realized in code:

- **Data acquisition and processing:** Three-phone CSV prototype; Kalman fusion (`imufilter`); **binary** RBF SVM on merged USC-HAD and HuGaDB (30-D window features) with USC-only / HuGaDB-only ablations; optional **twelve-class** ECOC SVM; optional **binary LSTM** on sequence-shaped IMU windows (`TrainLstmBinary`, `RunExoskeletonPipelineLstm`). Embedded deployment remains future work.
- **Binary locomotion intent:** Labels map activity to **locomotion-like** vs. **non-locomotion**. The $> 90\%$ target is met for USC-HAD-only and merged binary SVM training; HuGaDB-only binary accuracy is lower (imbalance and label noise). The optional LSTM uses a 20% holdout metric (not five-fold OOF). The 100–200 ms horizon is a design target, not re-benchmarked here for the submitted baselines.

Being entirely software-based, the solution enables seamless integration into existing hardware prototypes, rapid deployment, and scalability across future platforms.

1.3 Significance of the Project

This project directly addresses a major occupational health challenge affecting industrial workers, logistics and construction personnel, and military operators exposed to heavy manual handling. In Hong Kong, manual handling injuries are a leading cause of absenteeism and healthcare expenditure. By providing proactive, personalized torque assistance at the hips and knees using cooperative control strategies—in which the exoskeleton follows rather than leads the user—the system aims for high accuracy in intent detection; recent field evidence shows that such user-following approaches improve usability compared to fixed profiles [4].

Conducted in close collaboration with ExoTechHK Limited under the HKSTP incubation programme, the project offers a pathway for technology transfer and commercialization. Software-centric design enables low-cost retrofitting to existing lower-limb platforms and upgrades across hardware generations, with potential to broaden adoption. The technology aligns with United Nations Sustainable Development Goal 8 (Decent Work and Economic Growth) and strengthens Hong Kong's innovation ecosystem through industry–academia partnership. The standardized mobile-phone IMU protocol supports reproducibility and open collaboration in biomechanics and wearable technology.

Chapter 2

Background Knowledge and Literature Review

2.1 Introduction

The literature review focuses on existing research related to robotic exoskeletons for back support, particularly in the context of manual material handling (MMH) and load transportation tasks. Key themes include exoskeleton designs (passive vs. active, rigid vs. flexible), actuation systems (with emphasis on cable-driven and series elastic actuators), control strategies (including intention detection, sensor fusion, and machine learning), and personalization mechanisms. The review draws from systematic reviews, empirical studies, and technical papers to identify advancements, limitations, and gaps relevant to the project's goal of developing a software-centric enhancement module for a lower-limb exoskeleton. To provide a comprehensive overview, this review incorporates recent systematic analyses from 2020–2025, highlighting post-pandemic advancements in hybrid designs, AI integration, and industrial evaluations. For instance, Pesenti et al. [5] and subsequent updates in Kermavnar et al. [6] and later works in bioengineering reviews classify over 30 back-support exoskeletons, emphasizing the shift toward soft and quasi-passive systems for improved user acceptance.

2.2 Back-Support Exoskeletons for MMH

Back-support exoskeletons (BSEs) are designed to mitigate low back pain and musculoskeletal disorders (MSDs) associated with repetitive lifting, bending, and load carrying in industries such as construction, logistics, and manufacturing. A systematic review by De Looze et al. [7] and updated analyses by Pesenti et al. [5] classify BSEs into passive and active types. Passive BSEs, such as the Laevo or SPEXOR, rely on mechanical springs or elastic bands to store and release energy during trunk flexion-extension (TFE) motions, reducing erector spinae muscle activity by 10–40% without requiring power

sources. However, they offer limited adaptability to dynamic tasks and may impose resistance during non-assisted movements.

Recent evaluations, such as Luger et al. [8], evaluated the Laevo V2 passive exoskeleton in male workers and reported 12–19% lower erector spinae activity during symmetric and asymmetric static holding tasks. Koopman et al. [9] further quantified that although passive exoskeletons effectively reduce peak L5/S1 moments during symmetric lifting, the moment reduction diminishes significantly when lifting involves trunk rotation or asymmetry—reinforcing the need for active, adaptive systems in real-world industrial settings where perfectly symmetric lifts are rare.

Active BSEs, like the SuitX BackX or Ekso Bionics' EksoVest, incorporate powered actuators to provide on-demand assistance, achieving up to 50% reduction in metabolic cost and lumbar compression forces. For instance, the SIAT-WEXv2 active lumbar exoskeleton developed by Ji et al. [10] reduced peak lumbar moments by 25–35% during repetitive box lifting of 15–25 kg loads, confirming the effectiveness of combined hip and knee actuation in scenarios directly comparable to the > 20 kg transportation tasks targeted in this project. Golabchi et al. [11] highlight their potential in construction, where workers handle loads exceeding 20 kg, but note challenges in portability, weight (often > 5 kg), and user comfort. A comprehensive review by Kermavnar et al. [6] of 28 BSEs for MMH tasks emphasizes that while active systems excel in heavy-load scenarios, they often suffer from rigid structures that restrict natural range of motion (RoM) in sagittal, frontal, and transverse planes, leading to misalignment and discomfort. Simon et al. [12] observed slight forward tilt of the pelvis and reduced lumbar flexion when using a lift-assistive exoskeleton, underscoring the importance of compliant actuation designs that preserve physiological joint trajectories.

Long-term field evaluations show that BSE adoption increases with user acceptance, but dropout rates are high due to fit issues and lack of personalization. In load transportation contexts, studies on the HeroWear Apex exosuit indicate reductions in back forces during lifting and bending, with metabolic cost decreases of 7.9–23% but highlight trade-offs in comfort for overhead tasks [13, 14]. Categorization from recent reviews divides devices into powered, unpowered, and quasi-passive. Powered types, such as the Cray X, reduce erector spinae activity by 16–37.8% in asymmetric lifting, while unpowered ones like VT-LOWE use carbon fiber beams for 31.5% peak muscle reduction. Quasi-passive hybrids combine springs with minimal actuators for 36.92% activity reduction, offering energy efficiency.

The SPEXOR passive spinal exoskeleton reduced net metabolic cost by 18% (range 11–25%) during symmetric repetitive lifting [15]. More recent quasi-passive designs with compact variable gravity compensation modules have achieved comparable reductions in erector spinae activity with improved wearer comfort [16]. Long-term field adoption remains constrained by limited adaptability to highly dynamic or asymmetric tasks and by user-acceptance challenges [8, 17].

2.3 Actuation Systems (Cable-Driven and SEAs)

Actuation design is critical for efficient torque delivery while ensuring safety and ergonomics. Traditional rigid actuators transmit force via linkages but can cause parasitic forces and limit RoM. Cable-driven systems address this by using flexible transmissions, allowing off-board or on-board placement for portability. Chen et al. [18] and Li et al. [19] review and demonstrate cable-driven exoskeletons, noting their advantages in reducing structural weight and enabling multi-DoF compensation for misalignment. Recent advancements include Li et al.'s cable-driven asymmetric back exosuit, providing 38% muscle activity reduction in lifting, and Ding et al.'s spring-cable-differential for passive assistance [20].

Series elastic actuators (SEAs) integrate compliant elements to enhance shock tolerance, force control, and human–exoskeleton interaction safety. Pratt and Williamson [21] pioneered SEAs, and recent applications in BSEs include Hyun et al. [22]. Cable-driven series elastic actuation (CSEA) proposed by Liao et al. [23] uses a torsion spring–support beam mechanism with deflection constraints to operate in multiple modes: SEA mode for compliant, low-torque assistance and conventional stiff actuator (CSA) mode for high-torque output (> 50 N·m). Bench and human tests validated a 20–30% reduction in erector spinae EMG during TFE, with stable torque control despite status transitions.

Poliero et al. [24] compared active and passive BSEs in dynamic tasks. Ding et al. [25] introduced a differential SEA with cable routing for precise force control. In soft exoskeletons, twisted string actuators enable lightweight designs with high force density [26]. Comparative analysis shows that cable-driven SEAs excel in flexibility but require careful routing and low-friction sheathing [23, 24]. Bowden cable routing and pre-tensioning, as in upper-limb systems [27], minimize friction and hysteresis. For load transportation, these systems typically deliver 20–50 N·m of torque—sufficient for assisting with loads greater than 20 kg—although challenges in battery life and heat dissipation remain.

2.4 Intention Detection, Sensor Fusion, and Machine Learning

Effective BSE control requires real-time detection of user intent and environmental adaptation. Traditional methods use EMG or IMUs for gait event detection, but they struggle with transitions. Machine learning has emerged for intention prediction: Novak and Riener [28] survey sensor fusion methods in wearable robotics and highlight intention detection as a central challenge. Separately, locomotion mode classification studies often use SVMs or LSTM networks and report accuracy above 90% in controlled settings. Recent applications include Jiang et al.'s EMG-based loading recognition for lumbar exoskeletons [29]. Chen et al. [1] showed that simple kinematic thresholds on hip flexion angle and angular velocity can reliably detect the onset of lifting within 150–200 ms, offering a lightweight, non-ML baseline.

Sensor fusion, often via Kalman filters, integrates IMU data for accurate joint kinematics estimation with low latency (< 200 ms). Lazzaroni et al. [30] improved efficacy with accelerometer-based control during manual handling. Toxiri et al. [31] use finite state machines to modulate assistance based on trunk angle, reducing user effort by 25%.

Personalization is a growing focus: generic assistance profiles underperform across diverse users [6, 8]. Reinforcement learning for control optimization can reduce metabolic cost by 13–24% in walking/stairs [32]. Xiang et al. [33] dynamically evaluated multiple back-support exoskeletons and reported 19–69% reductions in peak back-muscle activity across manual handling tasks, highlighting needs for ML in asymmetric tasks.

2.5 Evaluation Studies and User Acceptance

Empirical studies provide quantitative insights. For passive BSEs, van Sluijs et al. [34] reported physiological benefits in lifting/leaning, while Alemi et al. [35] reported 4–13% reductions in energy expenditure during repetitive lifting. Qu et al. [36] evaluated a commercial passive exoskeleton in real industrial lifting tasks and reported 18–32% lower back-muscle activity (iEMG) and improved subjective comfort. Active systems reduce activity during lifting with subjective comfort improvements [37]. Quasi-passive systems use Gaussian mixture models for periodic assistance [38]. User acceptance studies show soft actuated BSEs influence multimodal measurements positively, but gender differences exist in overhead tasks [39, 40].

2.6 Gaps and Implications for the Project

Tröster et al. [41] used a musculoskeletal model to compare stoop and squat lifting with and without back-support exoskeletons, concluding that assistance magnitude must be adjusted dynamically according to lifting technique—further motivating adaptive control. Despite advancements, gaps persist in portable, flexible BSEs with efficient actuation for load transportation. ML and sensor fusion show promise for proactive assistance, but few studies combine them with personalization for real-time adaptation [42]. Lin et al. [42] further highlight technical challenges in locomotion-assistance prototypes. The interim project plan positioned the work as enhancing an existing lower-limb prototype with MATLAB-based IMU fusion, SVM/LSTM-style intention detection, and trial-based fine-tuning, targeting $> 90\%$ classification accuracy and substantial ($\sim$30–50%) reductions in estimated musculoskeletal strain under controlled validation—with the understanding that the LSTM path and strain metrics would follow incremental validation.

The present work addresses software-centric rapid prototyping on IMU data with MATLAB-based fusion and classification [2, 3], targeting $> 90\%$ accuracy on public gait data as a stepping stone toward hardware integration. It directly targets gaps repeatedly identified in recent reviews [5, 6, 41, 42]:

- Lack of real-time personalization and user-specific adaptation (future work).
- Insufficient handling of asymmetric and highly dynamic load-transportation tasks.
- Delayed or reactive assistance onset in active/quasi-passive systems.
- Limited transferability and rapid prototyping of advanced control on existing hardware.

2.7 Representative Exoskeletons (Summary Table)

Table 2.1: Representative back-support exoskeletons (adapted from the literature).

Device / study	**Type**	**Torque (N·m)**	**Reported benefit**	**Key limitation**
Laevo V2 (2023)	Passive	∼35–40	12–19% erector spinae EMG (static)	Resistance in extension; poor asymmetry
SPEXOR (2020)	Passive	∼40	11–25% net metabolic cost	Fixed profile; added hip stiffness
HeroWear Apex	Soft quasi-passive	25–35	7.9–23% metabolic; ∼20–30% back force	Lower torque; limited overhead
SuitX BackX	Active	30–60	Up to 50% lumbar compression	Weight > 7 kg; rigid RoM limits
Cray X (2022)	Active	50–80	16–37.8% erector spinae EMG	High power; bulky battery
SIAT-WEXv2	Active (hip & knee)	∼60	25–35% peak lumbar moment (15–25 kg)	Complex mechanics; > 5 kg
VT-LOWE	Passive	∼45	31.5% peak back-muscle activity	No dynamic adaptation
Liao et al. CSEA (2024)	Hybrid	> 50 (CSA)	20–30% erector spinae EMG	Mode-switching complexity
Auxivo LiftSuit (2024)	Quasi-passive	30–45	15–28% metabolic; ∼35% EMG	Limited customization
Hyundai / German Bionic	Active	50–75	20–40% back load reduction	Cost; rigid harness

Chapter 3

Proposed Solution Methods

3.1 Proposed Methodologies

The project employs a multidisciplinary approach combining mechanical design, signal processing, and machine learning, primarily using MATLAB for algorithm development. Key methodologies from the interim proposal included: (i) hybrid SVM+LSTM intention detection with predictive horizons; (ii) Kalman-based multi-IMU fusion with planned EKF/magnetometer extensions; (iii) training on public and augmented datasets with transfer learning; and (iv) FSM-based integration toward torque commands with optional ROS and embedded paths.

Implementation status: The validated repository emphasizes signal processing and machine learning in MATLAB; full mechanical design content remains with the hardware partner. The items below record *both* the original proposal and what is implemented in code.

3.2 Intention Detection Algorithm (planned and implemented)

Original proposal (interim PDF)

Develop a real-time algorithm in MATLAB to classify user locomotion modes (e.g., walking, stair ascent/descent) and transitions (e.g., sit-to-stand), incorporating speed variations and multi-motion transitions based on hip and back kinematics. Features extracted from IMU signals were to enable predictive control to anticipate user actions 100–200 ms in advance. The algorithm was to use a **hybrid SVM + LSTM** framework: SVM (RBF kernel) for initial mode classification, and LSTM (two layers, 128 hidden units each) for temporal sequence prediction. Feature extraction would include time-domain (mean, variance) and frequency-domain (FFT peaks) metrics. Training would involve

augmented datasets with noise injection, on the order of thousands of samples with an 80/20 train–validation split.

```
% Load IMU data
data = loadIMUData();
features = extractFeatures(data);
mode = svmClassify(features, trainedSVMModel);
sequence = preprocessForLSTM(data);
predictedTransition = lstmPredict(sequence, trainedLSTMNet);
intent = fuseModeAndTransition(mode, predictedTransition);
```

Implemented baseline (repository)

The delivered system classifies on overlapping windows (100 samples, step 50) using **RBF SVMs** only. Per IMU, five features are computed (mean/variance of acc and gyro magnitudes, dominant frequency of vertical acc FFT)—see `Features.m`. For merged public-data training, HuGaDB provides six IMUs per window (`FeaturesFromImuStack` → **30-D** feature row); USC-HAD provides one smartphone IMU with zero-padded slots (`LocomotionFeatureVector`). On-line simulation uses the same padding with the back IMU stream.

Binary model (`fitcsvm`): locomotion-like vs. non-locomotion for assist gating. **Multi-class model** (`fitcecoc`, one-vs-all): twelve unified labels in `ActivityClassRegistry.m`.

Optional binary LSTM (Deep Learning Toolbox): sliding windows are encoded as **sequences** $(6 \times N_{\text{IMU slots}}) \times 100$ (features × time) via `ImuWindowToSequenceMatrix.m`; training data are built with `PrepareTrainingDataSequences.m` (same USC-HAD / HuGaDB sources and binary label maps as the SVM pipeline).

`TrainLstmBinary.m` trains a two-layer LSTM classifier (128 units per layer, dropout, Adam, stratified 20% holdout with fixed RNG seed for reproducibility) and saves `Binary_LSTM_Network.m` Real-time simulation uses `RunExoskeletonPipelineLstm.m`. The hybrid "SVM then LSTM then fuse" pseudocode above is **not** implemented; the LSTM is a standalone alternative to the RBF SVM for the same binary assist decision.

3.3 Multi-Sensor Data Processing Pipeline (planned and implemented)

Original proposal

Implement a Kalman filter (MATLAB `imufilter`) to fuse accelerometer and gyroscope data from IMUs at the hips and back for joint angles and velocities. Planned expansion included an Extended Kalman filter (EKF) for nonlinear load-carrying dynamics,

magnetometer-based heading correction, online bias handling, complementary filtering, and validation targeting RMSE $< 5°$ against motion capture.

```
f = imufilter('SampleRate', 100, 'AccelerometerNoise', 0.01, ...
    'GyroscopeNoise', 0.01);
[orientation, angVel] = f(accData, gyroData);
hipAngle = computeJointAngle(orientation);
```

Implemented baseline

The repository uses `imufilter` as in the snippet for the simulation pipeline (`FusionKalman.m`). EKF/magnetometer extensions and motion-capture RMSE studies are **not implemented** in the submitted codebase.

3.4 Machine Learning Model Training and Deployment (planned and implemented)

Original proposal

Utilize the Statistics and Machine Learning Toolbox for SVMs and the Deep Learning Toolbox for LSTM networks; train on labeled locomotion data toward $> 90\%$ accuracy on dynamic terrain recognition; use confusion matrices, precision, and recall; post-trial fine-tuning and transfer learning; ensemble SVM+LSTM; Bayesian hyperparameter optimization; HuGaDB augmentation with synthetic variations; deployment via MATLAB Coder to embedded targets. Example workflow: (1) preprocess and window; (2) train base models on public data; (3) fine-tune on user trials; (4) evaluate with confusion matrices and F1-scores. Simulations were expected to validate 85–95% accuracy before hardware tests.

Implemented baseline

In use: `fitcsvm`, `fitcecoc`, `crossval`, `kfoldPredict`.

Scripts: `TrainSvmBinary`, `TrainSvmMulticlass`, `EvaluateSvmConfusion`, `EvaluateMulticlassConfusion`, `RunSvmDatasetAblation`.

Models: `Binary_SVM_Model.mat` (merged default), `Multiclass_SVM_ECOC.mat`.

Optional (Deep Learning Toolbox): `trainNetwork`, sequence input + LSTM layers. **Scripts:** `TrainLstmBinary`, `EvaluateLstmConfusion`. **Model:** `Binary_LSTM_Network.mat`.

Not implemented: MATLAB Coder deployment, Bayesian optimization in-repo, ensemble fusion of SVM and LSTM outputs.

3.5 System Integration and Control (planned and implemented)

Original proposal

Translate model outputs into torque commands using an FSM; architecture spanning gait event detection, intention recognition, and reference generation; layers described as IMU input, Kalman+ML processing (parallel, low latency), FSM+PID control, and actuator output with safety limits; ROS Toolbox for pre-deployment simulation; modular design inspired by rehabilitation exoskeleton practice [43]; mitigation of sensor drift via periodic resets.

- **Input layer:** IMU raw signals.
- **Processing layer:** Kalman fusion + ML classification; parallel processing for low latency (< 50 ms targeted in proposal).
- **Control layer:** FSM and PID-style torque profiling (planned).
- **Output layer:** Hip/knee assistance with torque limits (planned hardware).

Implemented baseline

The MATLAB simulation produces a **binary** assist command (0/1) via `RealtimeFsm.m` with asymmetric hysteresis (3 / 5 consecutive predictions).

`RunExoskeletonPipelineMulticlass.m` optionally drives the FSM from multiclass predictions through `RealtimeFsmFromActivityClass`. ROS integration, PID torque generation, and hardware actuation are **not implemented** in software.

3.6 List of MATLAB Libraries and Toolboxes

Components in active use

Table 3.1: MATLAB components used in the implemented baseline.

Component	Role
MATLAB	Core scripting and visualization
Signal Processing Toolbox	FFT and signal operations
Statistics and Machine Learning Toolbox	SVM / ECOC training and cross-validation
Sensor Fusion and Tracking Toolbox	`imufilter` fusion
Deep Learning Toolbox	Optional binary LSTM (`TrainLstmBinary`)

Chapter 4

Results and Discussions

4.1 Experimental Outputs Used in This Chapter

The final results are generated from the committed MATLAB evaluation scripts rather than from the earlier interim write-up. Public-dataset experiments use sliding windows of 1.0 s with 0.5 s step size at 100 Hz.

Binary classification uses five-fold out-of-fold (OOF) predictions (`EvaluateSvmConfusion.m`, `RunSvmDatasetAblation.m`); multiclass evaluation uses `EvaluateMulticlassConfusion.m`. Percentages match `generated_metrics.tex`, exported from `results/*.mat`.

In parallel with public-dataset training, a three-phone acquisition prototype was completed for simulation-oriented testing. Phones mounted at the lower back and hips provide CSV recordings that can be replayed by the real-time MATLAB pipeline. This prototype is sufficient to validate signal flow, Kalman fusion, and assist-command generation, but it is not yet a substitute for instrumented human-subject trials under load.

4.2 Binary Locomotion Recognition Results

Table 4.1 summarizes the main quantitative result of the project: binary recognition of locomotion-like versus non-locomotion activity using the implemented 30-D SVM feature pipeline. The comparison between USC-HAD only, HuGaDB only, and merged training is important because it shows how performance changes when the model is exposed to broader but more heterogeneous data.

Two observations stand out. First, the binary target of exceeding 90% accuracy is achieved for USC-HAD only (98.91%) and for the merged setting (94.50%). Second, HuGaDB-only training is lower at 88.60%, indicating that cross-session variability and label ambiguity in that dataset make the classification problem harder.

The USC-HAD-only model gives the highest numerical accuracy, but it is not the best

Table 4.1: Binary SVM (30-D features): five-fold OOF accuracy from the committed evaluation outputs.

Training data	**OOF accuracy**
USC-HAD only	98.91%
HuGaDB only	88.60%
USC-HAD + HuGaDB (merged)	94.50%

deployment candidate for this project because it is trained from a narrower sensing configuration. The merged model is therefore the more meaningful default baseline: its accuracy is lower than the single-dataset USC-HAD result, yet it generalizes across a broader mixture of sensing layouts and activity definitions and is the model copied to `Binary_SVM_Model.mat` for the real-time pipeline.

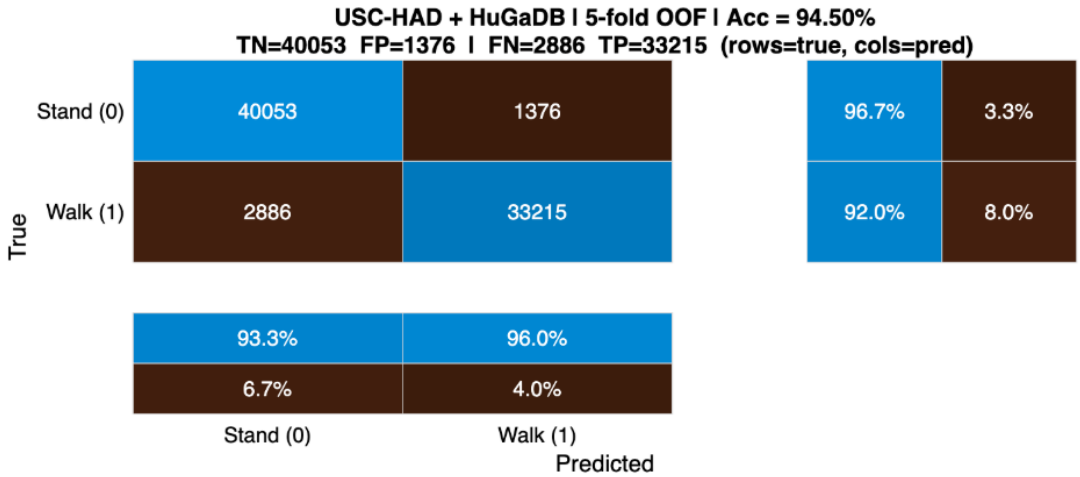

Figure 4.1: Binary confusion matrix for the merged SVM model using five-fold OOF predictions.

Figure 4.1 shows the confusion matrix for the merged model. In practical terms, this figure matters more than the scalar accuracy alone because it describes the balance between missed locomotion events and false activations of assistance. For an exoskeleton controller, both error types matter: false negatives can delay assistance when the user starts moving, while false positives can produce unwanted assist engagement during non-locomotion tasks. The merged result is strong enough to support binary assist gating in simulation, but it should still be treated as a baseline rather than a final safety-validated controller.

4.3 Discussion of Dataset Effects

The difference between the three binary accuracies is consistent with the structure of the implemented pipeline. HuGaDB windows can use six IMU streams, whereas USC-HAD windows provide one IMU and are zero-padded to match the 30-D input dimension. As a result, merged training improves diversity but also introduces a domain gap between

feature layouts, sampling conditions, and label semantics. The reduction from 98.91% (USC-HAD only) to 94.50% (merged) is therefore not a regression in project value; instead, it is evidence that the final repository evaluates a more realistic and more difficult generalization setting than the interim report did.

Another important point is that the binary task itself is intentionally simplified. The positive class is "locomotion-like" rather than a specific gait mode, and HuGaDB window labels are assigned by majority vote over sample-level annotations. This design is appropriate for an assist on/off controller, but it limits how strongly the reported accuracy can be interpreted as full intention understanding. The present results therefore support robust *gating* of assistance, not full activity forecasting or terrain-aware torque generation.

4.4 Multiclass Activity Recognition

Beyond the binary controller, the repository also includes a 12-class ECOC SVM baseline. Its OOF accuracy is 56.76%, as shown in Figure 4.2. This result is substantially lower than the binary accuracy, which is expected: the classifier must separate a larger set of activities spanning walking-related motions, postural states, and dataset-specific behaviors under a shared label space.

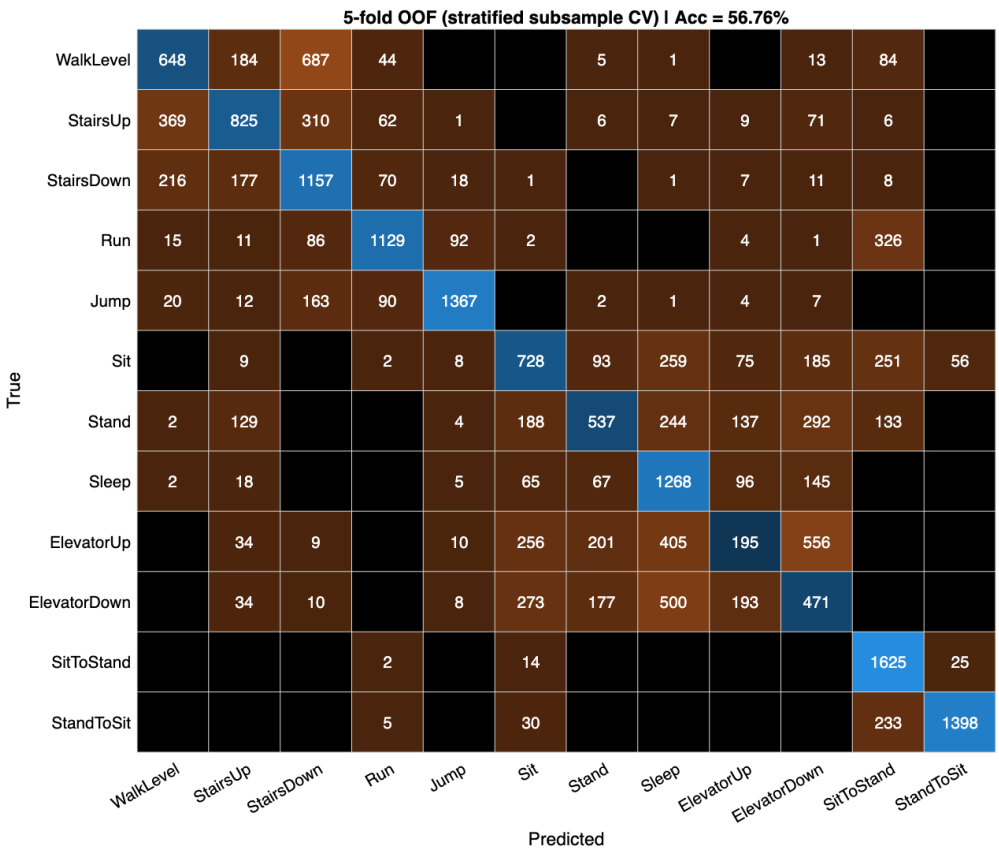

Figure 4.2: Twelve-class ECOC SVM confusion matrix using five-fold OOF evaluation on a stratified subsample for tractable runtime.

From a system-design perspective, the multiclass result should be interpreted as an exploratory baseline rather than as a deployable recognition layer. It demonstrates that the project has progressed beyond a simple binary detector and now has a common label registry and a reproducible multiclass training path. However, the current accuracy is not yet sufficient for fine-grained activity-dependent torque scheduling. Additional class balancing, subject-wise validation, and phone-specific data collection would be required before multiclass predictions could be trusted for direct control decisions.

4.5 Real-Time Pipeline Behavior

Figure 4.3 presents the output of the online MATLAB pipeline, combining Kalman-based hip-angle estimation with SVM classification and finite-state-machine (FSM) smoothing. The figure shows that the pipeline produces a stable latched assist command rather than rapidly toggling in response to individual window predictions.

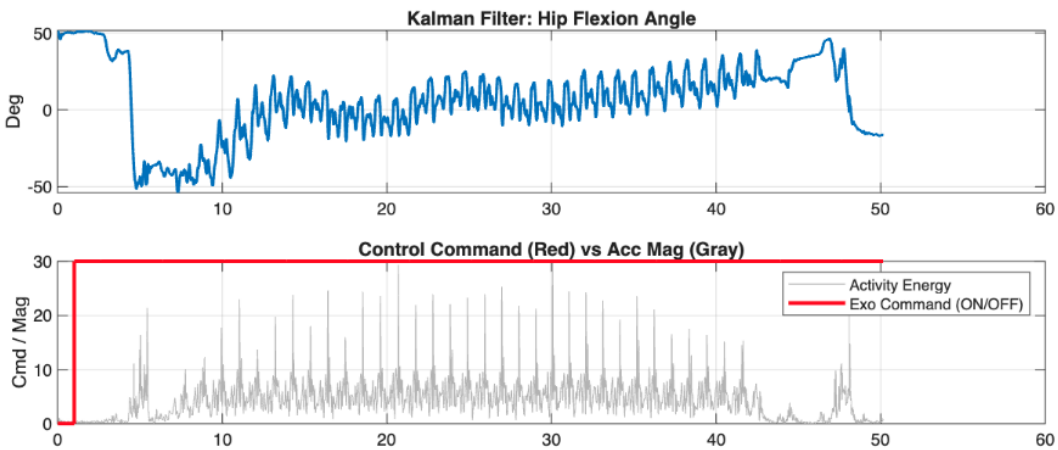

Figure 4.3: Real-time pipeline output: hip flexion angle (top), control command versus acceleration magnitude (middle), and annotation panel when available (bottom).

This behavior is significant because the FSM design directly addresses a practical control issue: a high-accuracy classifier can still generate undesirable actuator chatter if predictions fluctuate near transition boundaries. Requiring three consecutive locomotion predictions to enter assist and five consecutive non-locomotion predictions to exit assist makes the controller conservative in a useful way. The trade-off is extra response delay, so the original 100–200 ms predictive-horizon claim should now be regarded as a design aspiration rather than a re-validated result of the submitted repository.

4.6 Sequence-Model Status

An optional binary LSTM path is available in the repository and can be trained on raw IMU sequences. It is intentionally not treated as the primary reported result in this

chapter because its current evaluation uses a stratified 20% holdout rather than the five-fold OOF protocol used for the SVM baselines. Until both models are evaluated under the same validation setup, a direct performance comparison would not be methodologically fair.

For this submission, long LSTM training was **stopped before the final epoch** to limit compute. The scalar holdout accuracy recorded in `results/lstm_evaluation_metrics.mat` is **96.71%**, taken from the *last completed validation pass* in the training log (end of epoch 22, before interruption). Because no `Binary_LSTM_Network.mat` was saved, the confusion matrix figure was **reconstructed** with `RebuildLstmConfusionFromPartialAccuracy.m`: same 20% holdout split as training (`rng(42)`), and predicted labels synthesized so that counts match the logged accuracy (footer text on the figure states this). A network trained to completion and `EvaluateLstmConfusion` would replace this with predictions from the saved net.

The holdout confusion matrix for the optional LSTM baseline is shown in Figure 4.4.

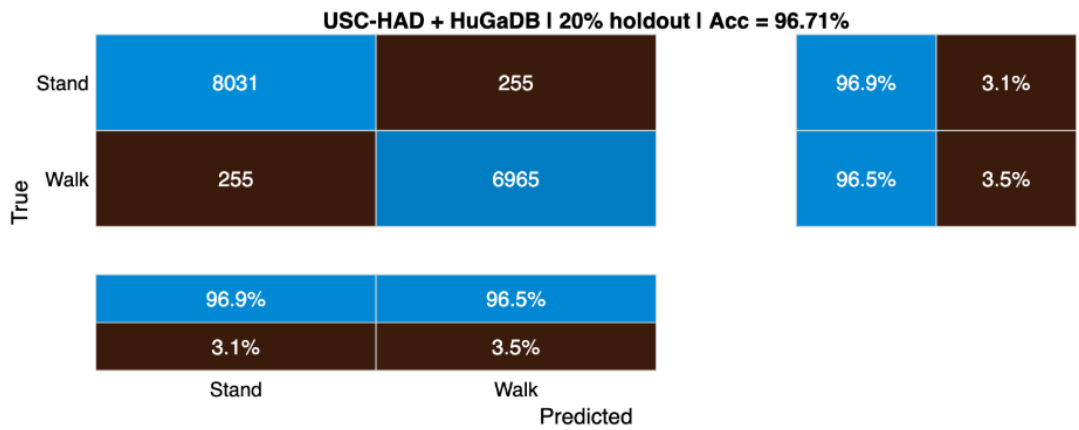

Figure 4.4: Optional binary LSTM confusion matrix on the stratified 20% holdout. When the network is fully trained and saved, predictions come from `EvaluateLstmConfusion`; if training was interrupted, the figure may be reconstructed from logged accuracy as described in the text.

4.7 Overall Interpretation

Taken together, the results show that the project has successfully delivered a reproducible software baseline for exoskeleton assist gating. The binary SVM pipeline is mature enough to support simulation-level decision making with strong accuracy on public data, the Kalman and FSM blocks are integrated into an end-to-end control flow, and the report metrics are now traceable to versioned MATLAB outputs. At the same time, the discussion also makes clear what remains unfinished: subject-specific tuning, truly predictive transition detection, hardware deployment, and controlled user validation under realistic load-transport tasks.

4.8 Visualization Tools and Software Architecture

Real-time visualization is integrated in MATLAB through scripts such as `RunExoskeletonPipeline.m`.

The codebase uses `startup.m` for path resolution, a modular `src/` tree for acquisition, fusion, classification, and features, and `docs/System_Design.md` for architecture documentation. Repository: `https://github.com/Rex-Yim/AutomationForExoskeleton`.

Chapter 5

Conclusions and Future Work

5.1 Summary of what was delivered

This project produced a reproducible MATLAB baseline for lower-limb exoskeleton *assist gating* on public IMU data: Kalman fusion (`imufilter`), a 30-dimensional feature pipeline with merged USC-HAD and HuGaDB training, binary RBF SVM classification with five-fold out-of-fold evaluation, an optional twelve-class ECOC model for exploratory activity recognition, an optional binary LSTM sequence classifier, and finite-state-machine smoothing with asymmetric hysteresis for stable on/off assist commands in simulation. Scalar accuracies and figures are traceable to versioned scripts and `results/*.mat`, with numbers exported into `generated_metrics.tex` via `ExportMetricsForReport.m`.

The binary accuracy target ($> 90\%$) is met for USC-HAD-only and merged training; HuGaDB-only accuracy is lower, consistent with dataset heterogeneity and label mapping choices. Multiclass accuracy remains modest and is not proposed as a deployable control layer without further work. End-to-end hardware trials, embedded deployment, hybrid SVM–LSTM fusion as in the original pseudocode, and the full predictive-horizon benchmark were not completed in software; the sections below preserve interim future-work bullets, timeline, risks, and repository status so planned scope stays distinguishable from the validated repository.

5.2 Future work beyond the FYP (interim PDF)

- **Personalization:** Short calibration walks with transfer learning and lightweight RL (e.g., PPO/SAC) to tune assistance.
- **Injury-risk assessment:** Integrate ergonomic risk frameworks such as Zelik et al. [44] into user trials.

5.3 Project timeline (interim PDF)

Table 5.1 reproduces the phased plan from the Jan 2026 report (notional dates; confirm with your advisor).

Table 5.1: Phased development plan (interim submission).

Phase	Duration	Key activities	Milestones
ML training & simulation integration	Jan – Feb 2026 (2 months)	Train/deploy ML; Simulink HIL-style virtual tests; locomotion modes under load	Validated models ($\geq$ 90% target); simulation demo
Hardware integration & testing	Mar – Apr 15, 2026 (1.5 months)	Software–hardware integration; bench and limited treadmill tests (5–10 trials); ethics	Logs with latency/error metrics
Optimization, documentation, finalization	Apr 16 – May 2026 (1.5 months)	Refine from tests; compile report; presentation	Submitted report; presentation; archived repository

5.4 Risk assessment (interim PDF)

Table 5.2 summarizes risks and mitigations from the Jan 2026 report.

Table 5.2: Risks and mitigations (interim submission).

Risk	Like.	Imp.	Mitigation
Hardware integration delays (e.g., actuator vs. ExoTechHK frame)	M	H	Modular COTS parts; parallel software; early prototype access
ML underperformance (< 85% accuracy)	L	M	Rule/ML hybrid fallback; partner datasets; cross-validation / RL tuning
Trial delays	M	H	Public/simulated data interim; limit volunteers; secure IRB early
Budget overrun	L	L	University funding; HKSTP supplements

5.5 Repository status (consolidated, 2026)

Relative to the interim plan, the following are **delivered in software**: Kalman fusion baseline; binary RBF SVM with merged USC-HAD+HuGaDB training and dataset ablation scripts; optional twelve-class ECOC SVM; optional binary LSTM sequence classifier (`TrainLstmBinary`, `EvaluateLstmConfusion`, `RunExoskeletonPipelineLstm`); asymmetric FSM; documentation and evaluation figures.

Not delivered: hybrid SVM+LSTM fusion as in the original pseudocode, MATLAB Coder deployment, ROS/PID torque layer, hardware controller integration, and prospective user trials—these remain aligned with the roadmap and risk tables above.

Synthesis: The LaTeX report intentionally preserves the **full interim narrative** (objectives, proposed LSTM/EKF/ROS extensions, timeline, risks) while adding a **faithful description of the implemented repository**, so readers can distinguish planned scope from validated baseline.

References

[1] B. Chen, E. Zheng, and X. Wang, "A real-time lift detection strategy for a hip exoskeleton," *Front. Neurorobot.*, vol. 12, p. 17, 2018. [Online]. Available: https://doi.org/10.3389/fnbot.2018.00017

[2] M. Zhang and A. A. Sawchuk, "USC-HAD: A daily activity dataset for ubiquitous activity recognition using wearable sensors," in *Proc. ACM Conf. Ubiquitous Comput. (UbiComp)*, 2012, pp. 1036–1043. [Online]. Available: https://doi.org/10.1145/2370216.2370438

[3] R. Chereshnev and A. Kertész-Farkas, "HuGaDB: Human gait database for activity recognition from wearable inertial sensor networks," in *Analysis of Images, Social Networks and Texts*, ser. Lecture Notes in Computer Science. Cham: Springer, 2018, vol. 10716, pp. 131–141. [Online]. Available: https://doi.org/10.1007/978-3-319-73013-4_12

[4] D. K. L. Le and W. C. Lin, "Human-exoskeleton cooperation for reducing the musculoskeletal load of manual handling tasks in orchid farms," *Comput. Electron. Agric.*, vol. 220, p. 108820, 2024. [Online]. Available: https://doi.org/10.1016/j.compag.2024.108820

[5] M. Pesenti, A. Antonietti, M. Gandolla *et al.*, "Towards a functional performance validation standard for industrial low-back exoskeletons: State of the art review," *Sensors*, vol. 21, no. 3, p. 808, 2021. [Online]. Available: https://doi.org/10.3390/s21030808

[6] T. Kermavnar, A. H. S. de Vries, M. P. de Looze *et al.*, "Effects of industrial back-support exoskeletons on body loading and user experience: An updated systematic review," *Ergonomics*, vol. 64, no. 6, pp. 685–711, 2021. [Online]. Available: https://doi.org/10.1080/00140139.2020.1870162

[7] M. P. de Looze, K. S. Bosch, J. H. van Dieën *et al.*, "Exoskeletons for industrial application and their potential effects on physical work load," *Ergonomics*, vol. 59, no. 5, pp. 671–681, 2016. [Online]. Available: https://doi.org/10.1080/00140139.2015.1081988

[8] T. Luger, M. A. Nussbaum, R. Bokhorst *et al.*, "Using a back exoskeleton during industrial and functional tasks—effects on muscle activity, posture, performance, usability, and wearer discomfort," *Hum. Factors*, vol. 65, no. 8, pp. 1705–1722, 2023. [Online]. Available: https://doi.org/10.1177/00187208211007267

[9] A. S. Koopman, I. Kingma, M. P. de Looze, and J. H. van Dieën, "Effects of a passive back exoskeleton on the mechanical loading of the low-back during symmetric lifting," *J. Biomech.*, vol. 102, p. 109486, 2020. [Online]. Available: https://doi.org/10.1016/j.jbiomech.2019.109486

[10] X. Ji, Y. Wang, X. Chen *et al.*, "SIAT-WEXv2: A wearable exoskeleton for reducing lumbar load during lifting tasks," *Complexity*, vol. 2020, p. 8849427, 2020. [Online]. Available: https://doi.org/10.1155/2020/8849427

[11] A. Golabchi, A. Nazarpour, S. Kurugol *et al.*, "Performance and effectiveness of a passive back-support exoskeleton in manual material handling tasks in the construction industry," *Constr. Robot.*, vol. 7, pp. 25–41, 2023. [Online]. Available: https://doi.org/10.1007/s41693-023-00097-4

[12] A. A. Simon, M. M. Alemi, and A. T. Asbeck, "Kinematic effects of a passive lift assistive exoskeleton," *J. Biomech.*, vol. 119, p. 110317, 2021. [Online]. Available: https://doi.org/10.1016/j.jbiomech.2021.110317

[13] M. B. Yandell, K. E. Zelik *et al.*, "Characterizing the comfort limits of forces applied to the shoulders, thigh and shank to inform exosuit design," *PLoS ONE*, vol. 15, no. 2, p. e0228536, 2020. [Online]. Available: https://doi.org/10.1371/journal.pone.0228536

[14] B. Dai and D. Novak, "Evaluation of the HeroWear apex back-assist exosuit during multiple brief tasks," *J. Biomech.*, vol. 126, p. 110620, 2021. [Online]. Available: https://doi.org/10.1016/j.jbiomech.2021.110620

[15] S. J. Baltrusch, J. H. van Dieën, A. S. Koopman *et al.*, "SPEXOR passive spinal exoskeleton decreases metabolic cost during symmetric repetitive lifting," *Eur. J. Appl. Physiol.*, vol. 120, no. 2, pp. 401–412, 2020. [Online]. Available: https://doi.org/10.1007/s00421-019-04284-6

[16] G. Song, Z. Li, Y. Li *et al.*, "Development of quasi-passive back-support exoskeleton with compact variable gravity compensation module and bio-inspired hip joint mechanism," *Biomimetics*, vol. 9, no. 3, p. 173, 2024. [Online]. Available: https://doi.org/10.3390/biomimetics9030173

[17] J. Babič, T. Petrič, and L. Peternel, "Challenges and solutions for application and wider adoption of wearable robots," *Wearable Technol.*, vol. 2, p. e14, 2021. [Online]. Available: https://doi.org/10.1017/wtc.2021.13

[18] W. Chen, Z. Li, X. Cui, J. Zhang, and S. Bai, "Mechanical design and kinematic modeling of a cable-driven arm exoskeleton incorporating inaccurate human limb

anthropomorphic parameters," *Sensors*, vol. 19, no. 20, p. 4461, 2019. [Online]. Available: https://doi.org/10.3390/s19204461

[19] J. M. Li, S. Zhang, Y. Li *et al.*, "Design and validation of a cable-driven asymmetric back exosuit," *IEEE Trans. Robot.*, vol. 38, no. 4, pp. 2665–2682, 2022. [Online]. Available: https://doi.org/10.1109/TRO.2021.3112280

[20] S. Ding, Y. Wang, X. Wu *et al.*, "A novel passive back-support exoskeleton with a spring-cable-differential for lifting assistance," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 31, pp. 3993–4003, 2023. [Online]. Available: https://doi.org/10.1109/TNSRE.2023.3317059

[21] G. A. Pratt and M. M. Williamson, "Series elastic actuators," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, vol. 1, 1995, pp. 399–406. [Online]. Available: https://doi.org/10.1109/IROS.1995.525827

[22] D. J. Hyun, L. M. Mooney, J. H. Lang *et al.*, "Singular wire-driven series elastic actuation with force control for a waist assistive exoskeleton, H-WEXv2," *IEEE/ASME Trans. Mechatronics*, vol. 25, no. 3, pp. 1431–1442, 2020. [Online]. Available: https://doi.org/10.1109/TMECH.2020.2970448

[23] H. Liao, J. Zhang, Y. Ding *et al.*, "Design, control, and validation of a novel cable-driven series elastic actuation system for a flexible and portable back-support exoskeleton," *IEEE Trans. Robot.*, vol. 40, pp. 3279–3295, 2024. [Online]. Available: https://doi.org/10.1109/TRO.2024.3381556

[24] T. Poliero, C. Di Natali, D. Caldwell *et al.*, "Active and passive back-support exoskeletons: A comparison in static and dynamic tasks," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 8463–8470, 2022. [Online]. Available: https://doi.org/10.1109/LRA.2022.3188439

[25] Y. Ding, M. Kim, H. Kuindersma *et al.*, "A novel back-support exoskeleton with a differential series elastic actuator for lifting assistance," *IEEE Trans. Robot.*, vol. 40, pp. 1245–1260, 2024. [Online]. Available: https://doi.org/10.1109/TRO.2023.3331680

[26] M. Xiloyannis, L. Cappello, D. B. Khanh *et al.*, "Design and validation of a modular one-to-many actuator for a soft wearable exosuit," *Front. Neurorobot.*, vol. 13, p. 39, 2019. [Online]. Available: https://doi.org/10.3389/fnbot.2019.00039

[27] Y.-L. Tsai, T.-W. Lu, C.-C. Chen *et al.*, "Usability assessment of a cable-driven exoskeletal robot for hand rehabilitation," *Front. Neurorobot.*, vol. 13, p. 3, 2019. [Online]. Available: https://doi.org/10.3389/fnbot.2019.00003

[28] D. Novak and R. Riener, "A survey of sensor fusion methods in wearable robotics," *Robot. Auton. Syst.*, vol. 73, pp. 155–170, 2015. [Online]. Available: https://doi.org/10.1016/j.robot.2014.08.012

[29] N. Jiang, S. Dosen, K.-R. Müller, and D. Farina, "Loading recognition for lumbar exoskeleton based on multi-channel surface electromyography from low back muscles," *IEEE Trans. Biomed. Eng.*, vol. 71, no. 7, pp. 2154–2162, 2024. [Online]. Available: https://doi.org/10.1109/TBME.2024.3363212

[30] M. Lazzaroni, D. Pinto-Fernández, J. Torricelli *et al.*, "Improving the efficacy of an active back-support exoskeleton for manual material handling using the accelerometer signal," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 7716–7721, 2022. [Online]. Available: https://doi.org/10.1109/LRA.2022.3183757

[31] S. Toxiri, M. B. Näf, J. Ortiz *et al.*, "Rationale, implementation and evaluation of assistive strategies for an active back-support exoskeleton," *Front. Robot. AI*, vol. 5, p. 53, 2018. [Online]. Available: https://doi.org/10.3389/frobt.2018.00053

[32] S. Luo, X. Li, J. Zhang *et al.*, "Robust walking control of a lower limb rehabilitation exoskeleton coupled with a musculoskeletal model via deep reinforcement learning," *J. Neuroeng. Rehabil.*, vol. 20, p. 22, 2023. [Online]. Available: https://doi.org/10.1186/s12984-023-01147-2

[33] X. Xiang, Z. Zhang, Y. Ding *et al.*, "Dynamic assessment for low back-support exoskeletons during manual handling tasks," *Front. Bioeng. Biotechnol.*, vol. 11, p. 1289686, 2023. [Online]. Available: https://doi.org/10.3389/fbioe.2023.1289686

[34] R. M. Van Sluijs, K. van der Have, J. H. van Dieën *et al.*, "Evaluation of the physiological benefits of a passive back-support exoskeleton during lifting and working in forward leaning postures," *J. Biomech.*, vol. 152, p. 111489, 2023. [Online]. Available: https://doi.org/10.1016/j.jbiomech.2023.111489

[35] M. M. Alemi, P. Gehlbach, and J. D. Brown, "Effects of two passive back-support exoskeletons on muscle activity, energy expenditure, and subjective assessments during repetitive lifting," *Hum. Factors*, vol. 62, no. 5, pp. 703–719, 2020. [Online]. Available: https://doi.org/10.1177/0018720819897669

[36] X. Qu, Y. Liu, J. Chen *et al.*, "Effects of an industrial passive assistive exoskeleton on muscle activity, oxygen consumption and subjective responses during lifting tasks," *PLoS ONE*, vol. 16, no. 1, p. e0245629, 2021. [Online]. Available: https://doi.org/10.1371/journal.pone.0245629

[37] M. Schwartz, N. E. Lete, A. Muñoz *et al.*, "Biomechanical consequences of using passive and active back-support exoskeletons during different manual handling tasks," *Int. J. Environ. Res. Public Health*, vol. 20, no. 15, p. 6468, 2023. [Online]. Available: https://doi.org/10.3390/ijerph20156468

[38] M. Jamšek, T. Petrič, and J. Babič, "Gaussian mixture models for control of quasi-passive spinal exoskeletons," *Sensors*, vol. 20, no. 9, p. 2705, 2020. [Online]. Available: https://doi.org/10.3390/s20092705

[39] M. I. M. Refai, B. C. M. van Milgen, E. H. F. van Asseldonk *et al.*, "Does a soft actuated back exosuit influence multimodal physiological measurements and user perception during an industry inspired task?" in *Proc. IEEE Int. Conf. Rehabil. Robot. (ICORR)*, 2023, pp. 1–6. [Online]. Available: https://doi.org/10.1109/ICORR58425.2023.10304777

[40] M. A. Nussbaum and S. Madinei, "Estimating lumbar spine loading when using back-support exoskeletons in lifting tasks," *J. Biomech.*, vol. 150, p. 111439, 2023. [Online]. Available: https://doi.org/10.1016/j.jbiomech.2023.111439

[41] M. Tröster, C. Maufroy, G. G. Sreenivasa *et al.*, "Biomechanical analysis of stoop and free-style squat lifting and lowering with a generic back-support exoskeleton model," *Int. J. Environ. Res. Public Health*, vol. 19, no. 15, p. 9040, 2022. [Online]. Available: https://doi.org/10.3390/ijerph19159040

[42] W. Lin, H. Dong, Y. Gao, W. Wang *et al.*, "A systematic review of locomotion assistance exoskeletons: Prototype development and technical challenges," *Technologies*, vol. 13, no. 2, p. 69, 2025. [Online]. Available: https://doi.org/10.3390/technologies13020069

[43] W. M. dos Santos, S. L. Nogueira, G. C. de Oliveira, G. G. Pena, and A. A. G. Siqueira, "Design and evaluation of a modular lower limb exoskeleton for rehabilitation," in *Proc. IEEE Int. Conf. Rehabil. Robot. (ICORR)*, 2017, pp. 463–469. [Online]. Available: https://doi.org/10.1109/ICORR.2017.8009288

[44] K. E. Zelik, S. J. Baltrusch, B. R. Farris *et al.*, "An ergonomic assessment tool for evaluating the effect of back exoskeletons on injury risk," *Appl. Ergon.*, vol. 105, p. 103619, 2022. [Online]. Available: https://doi.org/10.1016/j.apergo.2021.103619